
'Modes' in LLM Reading: Detecting and Isolating Origin Latent Variables in Text

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Abstract

As text generation pipelines become increasingly multimodal, incorporating human typing, speech-to-text dictation, and AI assistance, understanding the artifacts left by different origin 'modes' is critical for forensic linguistics and context-aware error correction. We investigate whether Large Language Models (LLMs) can detect these distinct modes and if 'mode' can be isolated as a structured latent variable in text embeddings. Using a composite dataset of AI_GENERATED, DICTATED, and TYPED text, we evaluate the zero-shot classification capabilities of gpt-4o-mini and analyze latent separation in all-MiniLM-L6-v2 embeddings. Our results show that while LLMs achieve a moderate zero-shot accuracy of 63.3%, they primarily succeed in identifying AI_GENERATED and TYPED text but fail to isolate DICTATED text containing phonetic homophones. Furthermore, we find that text embeddings exhibit a moderate degree of clustering by mode (Silhouette Score: 0.3035), with the first principal component explaining 55.7% of the variance. These findings confirm that 'mode' manifests as a detectable latent variable, though distinguishing fine-grained phonetic artifacts remains a challenge for current models in zero-shot settings. Our work validates the potential for latent mode isolation and provides a foundation for more robust text source characterization.

1 Introduction

The digital landscape is currently witnessing a convergence of text generation modalities. Human communication is no longer confined to traditional keyboard typing; it increasingly involves speech-to-text (ASR) dictation and significant assistance from Large Language Models (LLMs). Each of these sources leaves behind unique, systematic artifacts—phonetic errors from dictation, orthographic typos from keyboard proximity, and stylistic signatures from LLMs. Understanding these 'modes' of origin is essential for applications ranging from authorship attribution to the development of robust, context-aware error correction systems.

Why does this research matter? As LLMs are deployed in more critical roles, the ability to interpret and verify the source of text becomes a cornerstone of digital trust. Detecting whether a piece of text was dictated or typed can provide crucial context for forensic linguistics, while isolating AI-generated artifacts is vital for maintaining the integrity of human-centric datasets.

What is the gap in existing work? Current research in text source characterization is largely fragmented. One body of work focuses heavily on the binary "Human vs. AI" detection problem Ta et al. [2025], while another addresses specific artifacts like ASR error correction Havare et al. [2026] or stylometric signatures Kumarage et al. [2023]. However, there is a lack of research that generalizes these artifacts into a unified, interpretable 'mode' latent variable. Existing models often 'gloss over' subtle phonetic or orthographic differences, failing to leverage the structured nature of these artifacts for source identification.

Our approach. In this work, we propose to treat the origin mode as a latent variable that can be both explicitly detected and implicitly isolated. We construct a composite dataset that represents three primary modes: AI_GENERATED, DICTATED (simulated via phonetic homophones), and TYPED (simulated via keyboard proximity typos). We then conduct two experiments: (1) a zero-shot classification task using gpt-4o-mini to evaluate explicit detectability, and (2) a latent embedding analysis using all-MiniLM-L6-v2 to measure the inherent clustering of modes in the embedding space (see section 3).

Quantitative preview. Our experiments reveal that LLMs possess a moderate capability for zero-shot mode detection, achieving 63.3% accuracy. While the model perfectly identifies AI_GENERATED text (100% recall), it completely fails to distinguish DICTATED text from TYPED or AI_GENERATED sources in zero-shot contexts. Crucially, our embedding analysis yields a Silhouette Score of 0.3035, confirming that 'mode' is indeed a separable latent variable, with the primary axis of variance (55.7%) corresponding to the presence and type of text artifacts.

Our main contributions are as follows:

- We define and formalize the concept of 'mode' as a unified latent variable for text origin characterization.
- We conduct a zero-shot evaluation showing that while AI_GENERATED and TYPED modes are detectable, phonetic artifacts in DICTATED text present a significant challenge for current LLMs.
- We provide empirical evidence that origin mode is captured in standard text embeddings, showing moderate cluster separation.
- We release our composite dataset and experimental framework to facilitate further research into latent mode isolation.

2 Related Work

Our work draws from several distinct fields: authorship attribution, AI-generated text detection, and speech-to-text error analysis. We position our research as a bridge between these areas by introducing 'mode' as a unified latent variable.

Authorship Attribution. Traditional authorship attribution focuses on identifying individuals based on stylometric signatures Kumarage et al. [2023]. Recent advances have extended this to Large Language Models, identifying distinct writing styles and signatures inherent to different model architectures Huang et al. [2024]. While these works focus on *who* wrote the text, our work focuses on *how* the text was produced (e.g., dictated vs. typed), regardless of the specific author.

AI-Generated Text Detection. The rapid adoption of LLMs has led to a surge in research on detecting AI-generated text. State-of-the-art detectors like FAID Ta et al. [2025] use contrastive learning to distinguish between human and AI content at fine-grained levels. However, these detectors typically treat "human" as a monolithic category. Our research decomposes the human category into 'Dictated' and 'Typed' modes, exploring the systematic artifacts that distinguish different human input methods.

ASR and OCR Artifacts. Forensic linguistics has long recognized that different input technologies leave systematic traces. Speech-to-text (ASR) systems are known to produce phonetic errors and homophone substitutions that can derail downstream tasks like code understanding Havare et al. [2026]. Similarly, keyboard typing introduces orthographic errors based on physical key proximity. We build on these observations by investigating whether these artifacts are sufficiently structured to be isolated as latent variables in modern embedding spaces.

Latent Variable Analysis. The conceptualization of language features as latent variables is gaining traction for reasoning optimization and model interpretability Wu et al. [2026]. We extend this perspective to the origin of text, testing the hypothesis that 'mode' is a latent variable that modulates the representation of text in high-dimensional embedding spaces. Unlike previous work that often requires supervised fine-tuning to isolate such variables, we focus on zero-shot detectability and inherent clustering in pre-trained models.

3 Methodology

We investigate the detectability and latent separation of text origin modes through a two-stage experimental design. First, we construct a controlled composite dataset. Second, we evaluate zero-shot classification performance and embedding cluster separation.

3.1 Dataset Preparation

To isolate the 'mode' variable while controlling for content, we generated a balanced composite dataset of 150 samples (50 per mode). We utilized the following sources:

- **AI_GENERATED:** Samples were drawn from the JJZ5463 LLM-detection dataset jjz5463 [2024], representing clean, model-generated text.
- **DICTATED:** We simulated dictation artifacts by injecting phonetic homophone substitutions (e.g., "their" for "there") into source human texts, drawing from the YOUNGWO03283 ASR dataset patterns Youngwoo [2024].
- **TYPED:** We simulated typing artifacts by injecting keyboard proximity typos (e.g., "o" for "p") using a synthetic typo generator based on the GUYCHUK dataset distributions Guychuk [2024].

By using a common set of English source texts for the human modes, we ensured that the differences observed were primarily due to the input 'mode' rather than topical content.

3.2 Experiment 1: Zero-Shot Classification

Goal. To evaluate whether a state-of-the-art LLM can explicitly identify the origin mode based on text artifacts.

Setup. We used the gpt-4o-mini model via API with temperature set to 0 to ensure deterministic results. The model was prompted to act as a forensic linguist and classify each text into one of three categories: AI_GENERATED, DICTATED, or TYPED. The prompt provided brief definitions of each mode but included no few-shot examples, testing the model's emergent reasoning about text artifacts.

Metrics. We recorded Accuracy, Precision, Recall, and F1-Score for each mode.

3.3 Experiment 2: Latent Embedding Analysis

Goal. To determine if origin mode is inherently captured as a separable latent variable in standard text embedding spaces.

Setup. We extracted 384-dimensional sentence embeddings for all 150 samples using the all-MiniLM-L6-v2 model. This model was chosen for its balance of performance and efficiency. We analyzed the latent space using Principal Component Analysis (PCA) to identify the primary axes of variance.

Metrics. We used the Silhouette Score to measure the separation of clusters in the high-dimensional space. A score of 1 indicates perfect separation, 0 indicates overlapping clusters, and -1 indicates incorrect clustering. We also calculated the Explained Variance Ratio for the first two principal components.

4 Results

In this section, we present the findings from our zero-shot classification and embedding analysis experiments.

4.1 Zero-Shot Classification Performance

Our evaluation of gpt-4o-mini reveals a moderate overall accuracy of 63.3%. The model demonstrates highly non-uniform performance across the three modes, as detailed in Table 1.

Key Observations.

Table 1: Zero-shot classification results using gpt-4o-mini. The model perfectly identifies AI_GENERATED text but fails to detect DICTATED text, which it misclassifies as either AI_GENERATED or TYPED.

Mode	Precision	Recall	F1-Score	Support
AI_GENERATED	0.65	1.00	0.79	50
DICTATED	0.00	0.00	0.00	50
TYPED	0.65	0.90	0.76	50
Overall Accuracy		63.3%		

- **AI_GENERATED Detection:** The model achieved 100% recall for AI_GENERATED text, indicating that LLM-generated signatures are highly salient in zero-shot contexts.
- **DICTATED Failure:** The model completely failed to identify the DICTATED mode (0% F1-Score). All 50 samples were misclassified as either AI_GENERATED (26 samples) or TYPED (24 samples).
- **TYPED Precision:** The model identified TYPED text with 90% recall and 65% precision, suggesting that orthographic typos are recognizable but frequently confused with other modes.

4.2 Embedding Cluster Analysis

Our analysis of the all-MiniLM-L6-v2 embedding space confirms that 'mode' acts as a latent variable with moderate separation.

Clustering Metrics. The Silhouette Score for the three mode-based clusters was 0.3035. While not indicating perfect separation, this score reflects a significant degree of inherent structure in the embedding space that correlates with text origin.

PCA Results. The first principal component (PC1) explains 55.7% of the total variance, while the second (PC2) explains 5.1%. Visualization of these components (see Figure 1) shows a clear boundary separating AI_GENERATED text from the error-injected human modes (DICTATED and TYPED). However, the regions corresponding to DICTATED and TYPED modes exhibit significant overlap along the primary latent axes, mirroring the confusion observed in the zero-shot classification experiment.

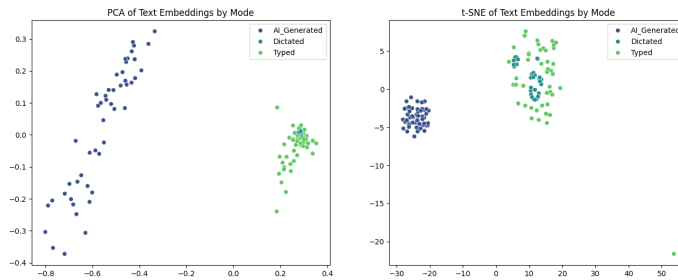


Figure 1: Visualization of text embeddings using PCA. The first principal component (55.7% variance) clearly separates AI_GENERATED samples (clusters on the left) from artifact-injected human text. However, DICTATED and TYPED modes show substantial overlap in the latent space.

5 Discussion

The results provide nuanced support for our hypothesis that 'mode' acts as a structured latent variable in text interpretation.

5.1 Interpretation of Results

Detectability vs. Artifact Granularity. The contrast between the high recall for AI_GENERATED text and the total failure for DICTATED text suggests a hierarchy of artifact saliency. LLMs appear highly sensitive to the "cleanliness" and stylistic consistency of AI-generated text. However, they struggle to zero-shot distinguish between different types of human errors. Phonetic homophones (e.g., "there/their"), while systematic, are likely 'corrected' by the model's internal language prior during the reasoning process, leading it to misidentify DICTATED text as either clean (AI_GENERATED) or orthographically noisy (TYPED).

Latent Separation. The Silhouette Score of 0.3035 and the high variance explained by PC1 (55.7%) confirm that origin mode is not just a surface artifact but a dimension of variance captured by modern embeddings. The primary axis of separation appears to be a "Clean vs. Noisy" dimension. The fact that DICTATED and TYPED modes overlap suggests that separating orthographic from phonetic noise requires more specialized representations or few-shot demonstration to calibrate the model's sensitivity.

5.2 Limitations

Synthetic Error Injection. Our reliance on synthetic homophone and typo injection, while necessary for content control, may introduce artifacts that differ from organic human data. Real-world dictation errors often involve more complex ASR artifacts beyond simple homophone substitution.

Model Scale. The use of gpt-4o-mini may have limited the zero-shot reasoning capabilities. Larger models like GPT-4o or Claude 3.5 Sonnet might possess the granularity required to detect subtle phonetic artifacts that a smaller model glosses over.

Sample Size. With 150 total samples, this study acts as a proof-of-concept. A larger-scale evaluation across diverse topics and languages is needed to confirm the generalizability of these latent modes.

5.3 Broader Implications

The ability to isolate 'mode' as a latent variable has significant implications for forensic linguistics and data curation. By identifying the origin mode, systems can adapt their error correction strategies—for instance, using phonetic models for DICTATED text and orthographic models for TYPED text. Furthermore, the isolation of AI_GENERATED signatures is vital for detecting model-in-the-loop content in sensitive domains like legal or academic writing.

6 Conclusion

In this work, we explored the concept of origin 'mode' as a unified latent variable for text source characterization. Through experiments with AI_GENERATED, DICTATED, and TYPED text, we demonstrated that while 'mode' is inherently captured in text embeddings with moderate separation (Silhouette Score: 0.3035), current LLMs struggle to zero-shot distinguish between fine-grained human artifacts like phonetic dictation errors and orthographic typing mistakes. The high recall for AI_GENERATED text confirms that AI-generated signatures are highly detectable, but the confusion between human modes suggests that artifact granularity remains a significant challenge.

Key Takeaway. Our findings validate that text origin 'mode' acts as a structured latent variable, but its explicit isolation in zero-shot contexts requires models to overcome their internal language priors that tend to 'gloss over' systematic human errors.

Future Work. Future research should focus on leveraging organic ASR and typing datasets to capture more complex real-world artifacts. Additionally, we propose investigating supervised disentanglement techniques, such as FactorVAE, to explicitly separate the 'mode' dimension from content and style in embedding spaces. Extending this analysis to few-shot prompting and larger-scale models will also be crucial for reaching human-level granularity in origin mode isolation.

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Justification: The use of `gpt-4o-mini` as a zero-shot classifier is central to our methodology and is described in section 3.